

TrES-3b

R-2696

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130611 · created 2026-08-31T23:49:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456454.7583 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.052
Transit depth (Rp/Rs)^2	3.75 +/- 2.0 [%]
Orbital Inclination (inc)	81.28 +/- 1.9
Airmass coefficient 1 (a1)	1.108 +/- 0.031
Airmass coefficient 2 (a2)	-0.089 +/- 0.027
Scatter in the residuals of the lightcurve fit is	2.81 %
Best Comparison Star	#2 - [499, 433]
Optimal Aperture	2.84
Optimal Annulus	12.97
Transit Duration (day)	0.048 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.194 $\pm$ 0.052 vs 0.16309 (deviation 0.39 σ)
Duration fit vs published	0.048 d vs 0.0567 d (deviation 0.26 σ)
Mid-time O-C	1.6 min
Depth SNR	2.4
Residual scatter	2.81 %

Light curve

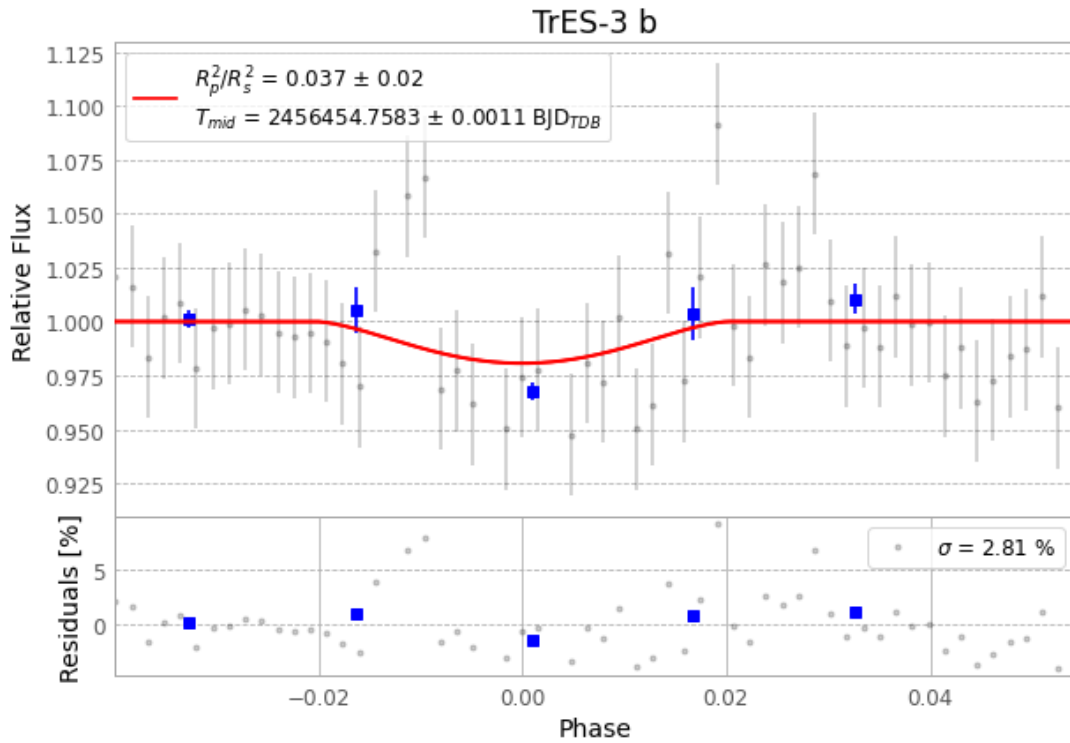


Figure 1 — FinalLightCurve_TrES-3 b_2013-06-10.png (SHA-256 1ad2c397859e3393...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [258, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2c57f0f200754031...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

60 FITS frames (39 MB), TRES-3130611045112.FITS → TRES-3130611074813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130611.log (0b2c1261a11f...), FinalLightCurve_TrES-3 b_2013-06-10.png (1ad2c397859e...), FinalLightCurve_TrES-3 b_2013-06-10.csv (250147b37b4a...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 23:49Z.